[NOT MEASUREMENT SENSITIVE]

A-A-59150A
March 12,. 1998
SUPERSEDING
A-A-59150
December 31, 1997

# COMMERCIAL ITEM DESCRIPTION

## CLEANING COMPOUND, SOLVENT, HYDROFLUOROETHER (HFE)

The General Services Administration has authorized
the use of this commercial item description by all federal agencies.

1. SCOPE. This commercial item description (CID) covers the requirements for a hydrofluoroether (HFE) cleaning compound solvent.

2. SALIENT CHARACTERISTICS.

2.1 Material. The HFE cleaning compound shall be a mixture of perfluoro-n-butyl methyl-ether (Chemical Abstract Service [CAS] Registry Number 163702-07-6) and perfluoroisobutyl methyl-ether (Chemical Abstract Service [CAS] Registry Number 163702-08-7).

2.2 Composition. The cleaning compound shall be a nonflammable, evaporating, non-residue solvent as specified in table I.

TABLE I. Composition and solvent purity.

| Component | Component purity (percent by weight) | Composition (percent by weight) |
|---|---|---|
| perfluoro-n-butyl-methyl-ether (CAS No. 163702-07-6) | see table II | 20 - 80 |
| perfluoroisobutyl-methyl-ether (CAS No. 163702-08-7) | see table II | 20 - 80 |

Beneficial comments, recommendations, additions, deletions, clarifications and any other data which may improve this document should be sent to: Commander, Naval Air Warfare Center Aircraft Division, Code 414100B120-3, Highway 547, Lakehurst, NJ 08733-5100.

FSC 6850

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2.3 Chemical and physical properties. The HFE solvent cleaning compound shall conform to the requirements of table II.

TABLE II. Chemical and physical properties.

| Property | Requirement | Test method or paragraph 1/ |
|---|---|---|
| Appearance | Clear, white water | 2.5 |
| Boiling point at 760 millimeters barometric pressure | 60.0 ±2.0°C (140.0 ±3.6°F) | ASTM-D1120 |
| Chemical purity | 99.5 percent by weight, minimum (min) | ASTM-D3447 |
| Impurities | Other halogenated solvents prohibited | ASTM-D3447 |
| Moisture content | 60 parts per million (ppm) by weight, maximum (max.) | ASTM-D3401 |
| Free fluoride ion | 1.0 ppm by weight, max. | ASTM-D3443 |
| Nonvolatile residue | 1.0 ppm, by weight, max. | ASTM-D2109 or ASTM-F331 |
| Particle matter<br>number of particles equal to or greater than 5.0 microns (µm);<br>number of particles greater than or equal to 1.0 µm<br>number of particles equal to or greater than 0.5 µm | <br>3 per milliliter (mL), max.<br><br>5 per mL, max.<br>16 per mL, max. | ASTM-F661 or SAE-ARP598 |

1/ Contractor test procedures equivalent to those specified may be substituted.

2.4 Shelf life. The HFE solvent cleaning compound specified in table I shall meet the specified requirements in table II after storage in unopened containers for not less than one year.

2.5 Appearance. The HFE solvent cleaning compound shall be a clear, colorless liquid (water white) with no evidence of particles. The appearance shall be determined by visually examining 500 ±50 mL of solvent for clarity, color, and solids in a clean, clear one-liter container. The inspection shall be performed by a person with normal visual acuity, natural or corrected, under bright light of not less than 100 foot-candles. A general purpose flashlight with 2 batteries and a lamp, positioned 18 inches perpendicular to a surface, will illuminate that surface with an intensity of not less than 100 foot-candles.

3. REGULATORY REQUIREMENTS.

Not applicable.



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## 4. QUALITY ASSURANCE PROVISIONS.

4.1 Product conformance. The products provided shall meet the salient characteristics of this CID, conform to the producer's own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial market. The Government reserves the right to require proof of such conformance.

4.1.1 Sampling and testing. Samples used for testing shall be taken from each lot, as defined by ISO8402, or other commercially accepted practice.

4.1.1.1 Sampling and testing before fill. Before filling each lot, the solvent shall be tested in accordance with table II for appearance, boiling point, chemical purity, impurities, moisture, free fluoride, and non-volatile residue. Sampling for particle matter shall be in accordance with 4.1.1.2.

4.1.1.2 Sampling and testing after fill. After filling each lot, the solvent shall be tested for particle matter as follows:

a. For 1- and 5-gallon containers, the first, last, and every 20th container shall be tested for particle matter in accordance with table II.

b. For 30- and 55-gallon containers, the first, last, and every other container shall be tested for particle matter in accordance with table II.

4.2 Market acceptability. The following market acceptability criteria are necessary to document the quality of the product to be provided under this CID.

4.2.1 The company furnishing the HFE cleaning compound solvent must have been producing a product meeting the requirements of this CID for at least 6 months.

4.2.2 The company must have offered the HFE cleaning compound solvent to a government laboratory for inspection.

## 5. PACKAGING.

5.1 Preservation, packing and marking. Preservation, packing, and marking shall be as specified in the contract or order and 5.1.1 through 5.1.3.



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5.1.1 Marking.

5.1.1.1 Safety warning labels. Containers shall be marked with a warning label addressing the following information:

a. Extreme conditions of heat (welding, open flame, misuse, or equipment failure) may produce hazardous decomposition products that include hydrogen fluoride. No smoking.

b. Use with adequate ventilation. In case of spills resulting in significant release in a confined space, warn others and leave immediately. Exposure to excessive vapor concentration in confined spaces may cause significant injury.

c. Avoid eye and prolonged skin contact. In case of contact with the eye, rinse immediately with plenty of water, and seek medical advice. Skin contact: flush skin with plenty of water. If irritation persists, call a physician.

d. For additional health and precautionary information, including air-monitoring methodology, refer to the material safety data sheet, or contact the manufacturer.

5.1.2 Material safety data sheets. A material safety data sheet shall be prepared and submitted in accordance with FED-STD-313, "Material Safety Data, Transportation Data And Disposal Data For Hazardous Materials Furnished To Government Activities."

5.1.3 Delivery options. The HFE solvent shall be delivered in 1-, 5-, 30-, or 55-gallon containers.

6. NOTES.

6.1 Intended use. This solvent is intended for precision cleaning applications.

6.2 Source of documents.

6.2.1 Federal Government activities may obtain copies of FED-STD-313 and DoD adopted non-government documents (NGSs) from established distribution points within their agencies; military activities may obtain these documents from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094. Other activities may obtain FED-STD-313 from the General Services Administration, Federal Supply Service, Specification Section, Suite 8100, 479 L'Enfant Plaza, SW, Washington, DC 20407 and NGSs from the NGS organization responsible for that document (see 6.2.2, 6.2.3, and 6.2.4).

6.2.2 The following documents may be obtained from the American Society for Testing and Materials, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959:



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| | | |
|---|---|---|
| ASTM-D1120 | - | Boiling Point of Engine Coolants, Standard Test Method for (DoD Adopted) |
| ASTM-D2109 | - | Nonvolatile Matter in Halogenated Organic Solvents and their Admixtures, Standard Test Method for (DoD Adopted) |
| ASTM-D3401 | - | Water in Halogenated Organic Solvents and their Admixtures, Standard Test Method for (DoD Adopted) |
| ASTM-D3443 | - | Chloride in Trichlorotrifluoroethane, Standard Test Method for (DoD Adopted) |
| ASTM-D3447 | - | Purity of Halogenated Organic Solvents, Standard Test Method for (DoD Adopted) |
| ASTM-F331 | - | Nonvolatile Residue of Halogenated Solvent Extract from Aerospace Components (Using Rotary Flash Evaporator), Standard Test Method for |
| ASTM-F661 | - | Particle Count and Size Distribution Measurement in Batch Samples for Filter Evaluation Using an Optical Particle Counter, Standard Practice for |

6.2.3 The following document may be obtained from SAE International, 400 Commonwealth Drive, Warrendale, PA 15096:

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)
AEROSPACE RECOMMENDED PRACTICE (ARP)

| | | |
|---|---|---|
| SAE-ARP598 | - | Aerospace Recommended Practice for the Determination of Particle Contamination in Liquids by the Particle Count Method (DoD Adopted) |

6.2.4 The International Standardization Organization (ISO) document ISO8402-94, "Quality Management and Quality Assurance - Vocabulary," which is DoD adopted, may be obtained from the American National Standards Institute (ANSI), 11 West 42nd Street, New York, NY 10036.

6.3 Government laboratory. Manufacturers wanting to have their cleaning solvent inspected may do so by contacting Commander, Attn.: Code 4.3.4.1, Building 2188, Naval Air Warfare Center Aircraft Division, Materials Lab, 22347 Cedar Point Road, Unit 6, Patuxent River, MD 20670-1161.

6.4 Purity and cleanliness of container. To conform to requirements of 2.4, all containers, including all caps, should be compatible with the solvent and precleaned before filling. Containers should be filled under environmental conditions and with equipment that will maintain the purity and cleanliness of the solvent. Glass containers should be water flushed and hot-air dried or rinsed with a solvent that is in accordance with this specification. All other containers should also be rinsed with a solvent that is in accordance with this specification before filling.



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6.5 Part Identification Number (PIN). The following part identification numbering procedure is for government purposes and does not constitute a requirement for the contractor. The PIN may be constructed as follows:

| AA59150 | - | XX |
|---|---|---|
| Document identifier | | Container size<br>01 = 1-gallon<br>05 = 5-gallon<br>30 = 30-gallon<br>55 = 55-gallon |

MILITARY INTERESTS:

Custodians:
Army - EA
Navy - AS
Air Force - 68

Review activity:
Navy - SH
DLA - GS
NASA - NA

Preparing Activity:
Navy - AS

(Project 6850-1244)



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